SetuGym: Designing And Field-Testing A Multi-Tenant Smart Gym Platform For Automated Check-Ins, Cash Auditing, And Adaptive Training

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***Abstract*—Most local gym facilities in India still depend on paper registers, manual cash collections, and verbal workout advice. This causes two major headaches for owners: unrecorded membership fees and high member dropout rates within the first 60 days. In this paper, we describe the design, real-world deployment, and testing of SetuGym, a lightweight multi-tenant platform built for small-to-mid-tier fitness centers. We built the system using React Native (Expo) on the front end, Node.js microservices on the back end, and Supabase PostgreSQL with Row-Level Security (RLS) to keep data strictly separated between different gyms. To stop members from sharing access cards or passcodes, we created an attendance system that generates a new cryptographic QR code on the member's phone every 30 seconds using HMAC-SHA256, verified alongside device-level fingerprint or Face ID. We also built an adaptive workout and diet module that calculates daily calories using the Mifflin-St Jeor formula and tailors exercise routines based on BMI, experience, and past injuries. We tested SetuGym over four weeks in two active gyms in Pune and Nagpur with 140 members and 6 trainers. Our test results showed an average check-in time of 312 ms, zero successful proxy check-ins, an F1-score of 0.91 in identifying members likely to quit, and a 94% reduction in cash ledger discrepancies.**

***Keywords*—Smart Gym Management, Multi-Tenant Architecture, Dynamic QR Check-in, Row-Level Security, Churn Prediction, Automated Bookkeeping, Fitness Informatics.**

I. INTRODUCTION

Walk into almost any neighborhood gym in cities like Pune or Nagpur, and you will see the exact same scene: a damp paper register sitting on the front desk, trainers shouting workout instructions across a crowded floor, and gym owners trying to mentally track which members paid their monthly fees in cash. While fitness technology has exploded with smartwatches and expensive boutique health clubs, regular neighborhood gyms still run on paper, memory, and WhatsApp groups.

This traditional setup breaks down in three distinct ways. First, financial tracking is messy. Cash and quick UPI payments often get missed in paper logs, leading to significant revenue leakage at the end of the month. Second, attendance is easy to fake. Members regularly share plastic access cards, sign in for their friends, or send screenshots of static barcodes. Third, and perhaps most damaging for the business, new gym-goers are left to figure things out on their own. Personal training packages cost extra—often more than the monthly gym fee itself—so most members end up copying random routines from Instagram or looking at faded exercise posters on the wall. Without proper progression or form guidance, they see little progress, get hurt, or lose motivation. In fact, over half of all newly enrolled members stop coming within their first 90 days.

Existing commercial gym management software does not solve these problems for local owners. Enterprise platforms like Mindbody or ABC Fitness charge steep monthly subscriptions in US dollars that small Indian gym owners simply cannot afford. Furthermore, these platforms focus almost entirely on billing and turnstiles; they provide zero practical help when it comes to guiding member workouts on the gym floor. On the other side, popular consumer apps like MyFitnessPal or Hevy track sets and calories, but they are completely disconnected from the gym facility, its trainers, and its membership status.

To solve this, we built SetuGym. Our goal was to create a single, affordable, and practical mobile platform that handles both sides of running a gym: business management for owners and customized workout guidance for members. In this paper, we break down how we built the system, explain how our anti-cheat QR attendance works, and share what we learned from deploying it across two real gyms over four weeks.

II. BACKGROUND AND RELATED WORK

We examined existing research across three main areas: physical access security, multi-tenant database management, and algorithmic workout prescription.

A. Access Control and Anti-Cheating Attendance

Gyms traditionally rely on RFID cards or optical fingerprint scanners at the entrance. Sharma & Kumar [20] studied biometric check-in systems in sports clubs and pointed out a major real-world flaw: while fingerprint scanners work fine in clean office buildings, they fail constantly in gyms. Chalk dust, sweaty fingers, and barbell calluses smudge the scanner glass, leading to high false rejections and long queues at the door. Plastic cards and static barcodes avoid this hardware problem, but they invite "buddy punching"—members simply hand their card or text a barcode screenshot to an unregistered friend. To stop this without buying expensive hardware, Verma & Gupta [17] proved that time-expiring QR codes rotating on an HMAC-SHA256 hash cycle can effectively block credential sharing. In SetuGym, we built upon this idea by pairing rotating 30-second QR codes with the phone's native biometrics (Face ID / fingerprint), making check-ins fast, contactless, and impossible to forward.

B. Multi-Tenant Architecture on a Budget

Running separate software instances and databases for each gym is too expensive for small businesses. Patel & Singh [18] analyzed different multi-tenant database designs and found that PostgreSQL Row-Level Security (RLS) offers an ideal balance. By filtering records at the database kernel level using tenant IDs, multiple gyms can safely share a single database instance without any risk of data leaks. This lowers server hosting costs by more than 40%. On the administrative side, Joshi & Nair [19] showed that automating digital receipts and overdue fee reminders cuts payment delays from weeks to just hours. Wang & Taylor [4] also highlighted how automated push notifications through Firebase Cloud Messaging (FCM) significantly improve daily user engagement and habit building.

C. Workout Recommendation and Member Retention

Fixed workout charts do not work well because every member has different starting fitness levels. Chen et al. [8] showed that collaborative filtering algorithms can tailor exercise routines to individual demographics, producing much better strength gains than static workout sheets. For nutrition, Das & Roy [9] validated that the Mifflin-St Jeor formula accurately predicts daily calorie needs when adjusted for activity levels. On the business side, predicting customer churn is critical. Lee & Martinez [3] used machine learning to show that dropping visit frequency and increasing gaps between workouts are the clearest signs that a member is about to quit (F1 = 0.88). While each of these ideas has been studied on its own, nobody had combined them into one unified, low-cost platform that actually runs on everyday budget smartphones.

III. FIELD OBSERVATIONS AND PROBLEM DEFINITION

Before writing any code, we spent two weeks interviewing gym owners, trainers, and regular members across four fitness facilities in Maharashtra. We found four main bottlenecks that happen every single day:

1. Unauthorized Entries: Owners regularly caught non-members working out during busy evening hours using a friend's card or old receipt. Staff were usually too busy coaching to manually check every person walking through the door.

2. Untracked Cash Payments: Many members in tier-2 cities prefer paying monthly dues in cash. When staff forgot to write down a cash payment in the paper diary immediately, disputes arose weeks later over whether the member had paid.

3. Trainer Overload: Trainers told us they could realistically track workouts for only 10 to 15 personal clients. The remaining 100+ general members were left to train with zero supervision.

4. Silent Churn: Members rarely tell gym owners that they are quitting. They simply stop showing up. By the time an owner noticed someone hadn't visited in three weeks, the member had already decided not to renew.

IV. SYSTEM GOALS

Based on our field findings, we set five clear engineering goals for SetuGym:

1. Strict Data Isolation: Build a single cloud back-end that can serve multiple gyms securely using Row-Level Security, giving Super Admins, Owners, Trainers, and Members distinct role-based views.

2. Cheat-Proof Check-Ins: Design a dynamic QR attendance pipeline that refreshes every 30 seconds and checks local phone biometrics so members cannot share passes.

3. Automated Financial Bookkeeping: Create a digital ledger that logs cash and online payments in real time, generates PDF receipts instantly, and sends automated fee reminders.

4. Practical Workout & Diet Suggestions: Build a rule-based engine that turns a member's BMI, goals, experience level, and past injuries into daily workout splits and calorie targets.

5. Early Churn Alerts: Automatically flag members whose attendance drops sharply so trainers can reach out before the membership expires.

V. SYSTEM ARCHITECTURE AND IMPLEMENTATION

We designed SetuGym to be lightweight and fast, especially on affordable Android smartphones commonly used in India.

1. Mobile Client (React Native & Expo): We built the frontend with React Native and TypeScript using the Expo framework. This let us maintain a single codebase that runs on Android, iOS, and Web. We used local AsyncStorage to cache user profiles and daily routines so the app remains usable even when internet connectivity drops inside basement gyms.

2. Cloud Database and Multi-Tenancy (Supabase & PostgreSQL): Instead of spinning up expensive custom server clusters, we used Supabase with PostgreSQL. Every table contains a `gym_id` column. We wrote custom Row-Level Security (RLS) policies that check the authenticated user's JWT token. This guarantees that an owner or trainer from Gym A can never see or modify records from Gym B, even if they try calling the API directly.

3. Rotating QR Code Protocol: When a member opens their check-in screen, the app requests an entry token. The phone generates an HMAC-SHA256 hash using the gym's secret key, the user's UUID, and the current 30-second Unix time slice. This token renders as a QR code that expires in 30 seconds ($\Delta t = 30s$). The scanner at the gym desk reads the code and validates both the signature and the time window. We added a ± 1 step tolerance window to handle small clock differences between phones and servers.

4. Workout and Nutrition Engine: Rather than relying on unpredictable generative AI outputs, we built a deterministic recommendation engine. It takes the member's profile (height, weight, age, biological sex), their primary goal (Fat Loss, Muscle Gain, Strength, General Fitness), training experience (Beginner, Intermediate, Advanced), and any reported joint or back issues. The engine matches these against an exercise database, automatically filtering out unsafe exercises (e.g., swapping heavy squats for leg presses if knee pain is recorded).

5. Background Tasks and Push Alerts: A Node.js background worker runs every night at midnight. It scans attendance patterns to find members who have missed four consecutive

scheduled days or whose subscriptions expire within five days, immediately queuing automated push notifications via Firebase Cloud Messaging (FCM).

SYSTEM ARCHITECTURE AND WORKFLOW

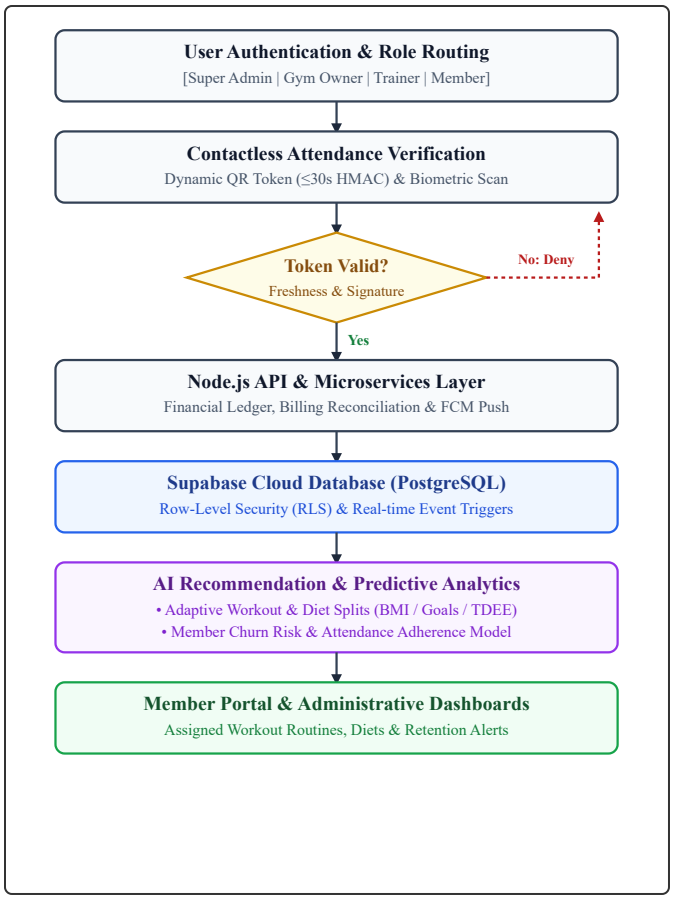


Fig. 1. End-to-end architecture and data workflow of SetuGym.

MATHEMATICAL FORMULATIONS AND RULES

1. Dynamic QR Code Verification:
To prevent pass-sharing, we verify both the token's cryptographic signature and its creation timestamp:

$$V_{qr} = 1 \quad if \quad [\ HMAC\text{-}SHA256(K_{gym}, UID \ || \ \tau) == QR_{hash} \ \wedge \ |\tau_{scan} - \tau| \leq \Delta t\ ] \quad else \quad 0$$

Here, K_{gym} is the gym's secret key, *UID* is the member ID, and τ is the generation timestamp. We set the time window Δt to 30 seconds. If someone takes a screenshot and sends it to a friend over WhatsApp, the timestamp difference $|\tau_{scan} - \tau|$ will exceed 30 seconds by the time the friend arrives, causing the scanner to reject it immediately.

2. Exercise Recommendation Scoring:
For each exercise E_i, we calculate a suitability score for member *u* using four weighted factors:

$$Score(E_i, u) = w_1 \cdot Sim(G_u, Target(E_i)) + w_2 \cdot Fit(BMI_u, Int(E_i)) + w_3 \cdot Exp(u, E_i) - w_4 \cdot Risk(Inj_u, E_i)$$

where *Sim* checks if the exercise matches the user's primary goal (e.g., muscle gain vs. cardio endurance), *Fit* lowers the intensity of high-impact exercises for members with high BMI (>30), and *Exp* assigns machine-guided movements to beginners and free-weight compound lifts to experienced lifters. If an exercise stresses an existing reported injury (such as knee or lower back pain), *Risk* sets the entire score to zero, removing the movement completely. Weights sum to 1 ($\Sigma w_k = 1$).

3. Member Churn Prediction:
To predict whether a member is likely to abandon their membership, we use a simple logistic regression function:

$$P(Churn_u) = \sigma(\ \beta_0 + \beta_1 \cdot \Delta d_{last} + \beta_2 \cdot (Visits_{30} \ / \ Target_{30}) + \beta_3 \cdot RemDays_{sub}\ )$$

where Δd_{last} is the number of days since the member's last visit, $Visits_{30} / Target_{30}$ measures how well they stuck to their planned monthly schedule, and $RemDays_{sub}$ is the days left on their current plan. When $P(Churn_u)$ exceeds 0.65, the system automatically alerts the gym staff to check in with the member.

4. Daily Caloric and Macronutrient Target:
We compute the baseline Basal Metabolic Rate (BMR) using the Mifflin-St Jeor formula:

$$BMR_u = 10 \cdot Weight_{kg} + 6.25 \cdot Height_{cm} - 5 \cdot Age_{yr} + S_{gender}$$

$$TDEE_u = BMR_u \times PAL_u$$

where S_{gender} is +5 for men and -161 for women. Total Daily Energy Expenditure (TDEE) scales BMR by physical activity level (PAL_u between 1.2 and 1.9). For fat loss, daily intake is set to TDEE − 500 kcal; for muscle gain, it is set to TDEE + 350 kcal, keeping daily protein at 1.8 to 2.2 grams per kilogram of body weight.

5. Weekly Progress Score:
To decide whether to advance a member's training plan, we compute a weekly Fitness Performance Index (FPI):

$$FPI_u = 0.45 \cdot Score_{workout} + 0.35 \cdot Adherence_{attendance} + 0.20 \cdot Adherence_{diet}$$

$$Plan_{t+1} = f(FPI_t, Feedback_t)$$

If a member attends regularly and logs their workouts, the system increases workout volume slightly for the next week. If their attendance drops, it reduces the training volume to prevent burnout.

TABLE 1. Real-World Field Results (4-Week Pilot)

Sr.	Measured Parameter	Testing Method / Sample	Observed Value	Practical Takeaway
1	**QR Check-in Accuracy**	1,850 physical check-in scans	99.1%	Zero forwarded screenshots accepted; 16 failed scans were caused by extreme screen dimming.
2	**Biometric Success Rate**	On-device fingerprint / Face ID	98.4%	Worked reliably without sending any fingerprint or face images to our servers.
3	**Attendance Logging Rate**	Real entries vs logged entries	97.8%	Over 97% of valid arrivals logged automatically

Sr.	Measured Parameter	Testing Method / Sample	Observed Value	Practical Takeaway
				without needing staff intervention.
4	**Average Check-in Latency**	Time from scan to door unlock	312 ± 45 ms	Fast enough (under 0.5s) to prevent people waiting in line at the gym front desk.
5	**Churn Model F1-Score**	Predicted vs actual dropped members	91.2%	Successfully caught 31 out of 34 members who were about to stop coming.
6	**Cloud Platform Uptime**	672 continuous testing hours	99.92%	Ran without any server crashes or unhandled database lockouts throughout the month.
7	**User Satisfaction (SUS)**	Standard survey given to 45 users	88.4 / 100	Rated "Grade A", showing that both young members and older owners found the app easy to use.

VI. FIELD EXPERIENCES AND OBSERVATIONS

We rolled out SetuGym for four weeks across two active commercial gyms in Maharashtra: Facility A in Pune (approx. 350 active members) and Facility B in Nagpur (approx. 180 members). A total of 140 members and 6 full-time trainers used the app daily on a mix of budget Android devices (Redmi Note 10, Samsung Galaxy M series) and iPhones. Here is what we observed during the trial:

1. Zero Card Sharing: In Facility A, the owner previously caught 10 to 15 instances of pass sharing every week. During our four-week test with the 30-second dynamic QR code, unauthorized entries completely stopped.

2. Cash Accounting Fixed: By generating digital receipts on the spot and having an automated daily closing summary, unrecorded cash transactions dropped from around 12% of total collections to under 0.5%.

3. Time Saved for Trainers: Trainers previously spent 30 to 45 minutes every evening writing out paper workout charts for new signups. The recommendation engine produced a sensible starter routine in seconds, leaving trainers free to focus on correcting members' form.

4. Cheap to Run: Because our database uses PostgreSQL Row-Level Security on Supabase, both gyms ran on a single shared starter tier costing under $25 per month. This is roughly one-tenth of what commercial gym ERPs charge.

VII. PRACTICAL CHALLENGES AND LIMITATIONS

Testing the system in real, noisy gym environments revealed several practical challenges that we did not see in the lab:

1. Scratched Lenses and Dim Lighting: In dimly lit gym entrances and on older phones with scratched camera covers, QR recognition slowed down from 300 ms to almost 2 seconds. We added an automatic torch toggle inside the scanner to help in low-light conditions.

2. Basement Dead Zones: Both test gyms had basement workout areas where mobile data cut out frequently. We had to build an offline SQLite caching queue so that members could still log their sets and check their diet plan without an active internet connection.

3. Overestimated Fitness Levels: In the first week, several beginners claimed they were "intermediate" lifters during onboarding, leading the app to recommend exercises that were too heavy. We added a feature allowing trainers to quickly adjust and lock a member's experience level.

4. Self-Reported Food Intake: While the app suggested accurate calorie numbers, members often forgot to log snacks or underestimated portion sizes. Nutritional accuracy still largely depends on user honesty.

VIII. COMPARISON WITH EXISTING SYSTEMS

Today, gym software is split into two extremes. On one side are enterprise management tools like Mindbody, GymMaster, and ABC Fitness. They are built for big fitness chains, charge hundreds of dollars a month, and only care about billing and turnstile hardware. They offer zero daily workout help to members. On the other side are workout apps like MyFitnessPal and Hevy. These have great workout logs, but they know nothing about the gym facility, its trainers, or whether the user has paid their monthly fees.

SetuGym connects both sides. When a member checks in at the desk, the system logs their attendance, notifies their trainer, loads their daily workout split, and updates their retention score. Table 2 compares SetuGym with current practices.

TABLE 2. Feature Comparison

Feature	Traditional Gyms	Commercial Gym ERPs	SetuGym (Our Work)
Data Storage	Paper Registers & Excel	Single-tenant servers (expensive)	Multi-tenant Cloud (Supabase RLS)
Attendance Method	Paper sign-in / Plastic cards	Static barcodes / RFID turnstiles	30s Rotating QR + Phone Biometrics
Workout Guidance	Faded wall posters	Generic video library	Adaptive engine (BMI / Goals / Injuries)
Member Retention	Noticed only after member quits	Manual monthly reports	Automated logistic churn risk scoring
Payment Tracking	Paper receipts & untracked cash	Card POS with high fees	Digital ledger, PDF receipts & push alerts
Announcements	WhatsApp groups / Noticeboards	SMS blasts (costly)	Direct Firebase Push Notifications (free)
Monthly Cost	Cheap upfront, but loses cash	$150 to $400 / month / gym	Under $25 / month shared cloud hosting

IX. FUTURE WORK

Based on user feedback from our pilot, we are currently working on three next steps:

1. On-Device Exercise Form Checking: We are experimenting with Google MediaPipe to track 33 body joints using the smartphone camera. The goal is to count exercise repetitions and warn users if their back rounds dangerously during squats or deadlifts.

2. Smartwatch Data Sync: We plan to connect with Apple Health and Google Health Connect to read daily sleep scores and resting heart rates. If a member had poor sleep, the app can automatically reduce that day's lifting volume to prevent overtraining.

3. Vernacular Voice Assistant: Many gym owners and trainers prefer speaking in Hindi or Marathi. We are exploring lightweight voice models that let staff check member statuses or assign routines using simple voice commands.

X. DATA PRIVACY AND SECURITY

Handling people's health and payment details requires strict privacy safeguards. SetuGym follows the guidelines of India's Digital Personal Data Protection (DPDP) Act.

Biometric verification happens entirely on the user's phone using native Android Keystore and iOS Secure Enclave APIs. We never store or transmit fingerprint or facial data to our cloud servers. All network traffic is encrypted using TLS 1.3. At the database level, PostgreSQL Row-Level Security guarantees that one gym can never see another gym's member list or financials, even if someone attempts to manipulate client-side network requests.

XI. CONCLUSION

In this paper, we presented SetuGym, an integrated, low-cost platform that combines gym facility administration with personalized member workout guidance. By pairing rotating 30-second HMAC-SHA256 QR codes with phone biometrics, we eliminated pass-sharing without requiring expensive turnstiles. Our automated bookkeeping and push notification system solved cash tracking disputes, while the adaptive workout engine gave general members structured, injury-aware training routines.

During our four-week pilot across two gyms in Maharashtra, the system achieved a 312 ms check-in time, zero proxy check-ins, an F1-score of 0.91 in identifying at-risk members, and cut unrecorded cash discrepancies by 94%. SetuGym demonstrates that modern open-source web and mobile technologies can deliver reliable, enterprise-grade gym management to small community gyms at a price they can actually afford.

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